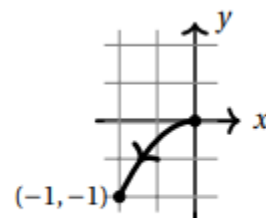
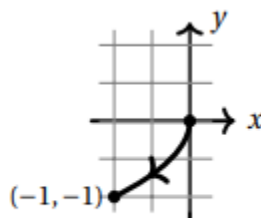
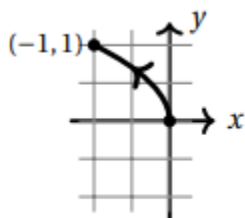
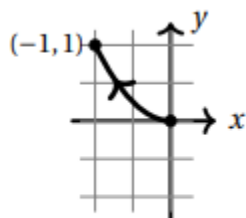


5.2.2 Line Integrals Worksheet

1. Let C be the curve in $\mathbb{R}^2$ parameterized by $\mathbf{r}(t) = \langle -t^2, t \rangle$ for $0 \leq t \leq 1$.

a) Mark the picture of C from among the choices below.



b) For the vector field $\mathbf{F} = \langle y, -x \rangle$ compute $\int_C \mathbf{F} \cdot d\mathbf{r}$.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \langle t, t^2 \rangle \cdot \langle -2t, 1 \rangle dt = \int_0^1 (-2t^2 + t^2) dt = \int_0^1 (-t^2) dt = -\frac{1}{3}t^3 \Big|_0^1 = -\frac{1}{3}$$

2. Find the work done by the force field $\mathbf{F}(x, y) = \langle y, x \rangle$ on a particle that moves along the cosine curve from $(0, 1)$ to $\left(\frac{\pi}{2}, 0\right)$.

We first parameterize the curve C we are moving along as $\mathbf{r}(t) = \langle t, \cos(t) \rangle$ where $0 \leq t \leq \frac{\pi}{2}$.

We then calculate

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{\pi/2} \langle \cos(t), t \rangle \cdot \langle -\sin(t), 1 \rangle dt = \int_0^{\pi/2} (t - \cos(t)\sin(t)) dt = \left(\frac{1}{2}t^2 - \frac{1}{2}\sin^2(t) \right) \Big|_0^{\pi/2} \\ &= \left(\frac{1}{2} \left(\frac{\pi}{2} \right)^2 - \frac{1}{2} \right) - 0 = \frac{\pi^2 - 1}{8} \end{aligned}$$

3. Three continuous vector fields $\mathbf{F}, \mathbf{G}, \mathbf{H}$ on $\mathbb{R}^2$ are plotted below in the region with $-1 < x < 1$ and $-1 < y < 1$. Circle the best response for each of the following questions.

- a) Which vector field represents $\vec{\nabla} f$, where $f(x, y) = xy$.

☒ **F** ☐ **G** ☐ **H**

We see that $\vec{\nabla} f = \langle y, x \rangle$ which is represented by **F**.

- b) A particle moves along a straight line from $(-1, -1)$ to $(1, 1)$. If the vector fields represent force fields, then:

The work done by **G** on the particle is...

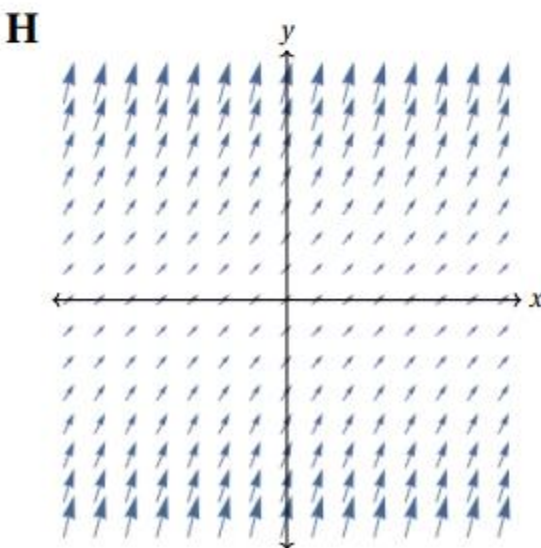
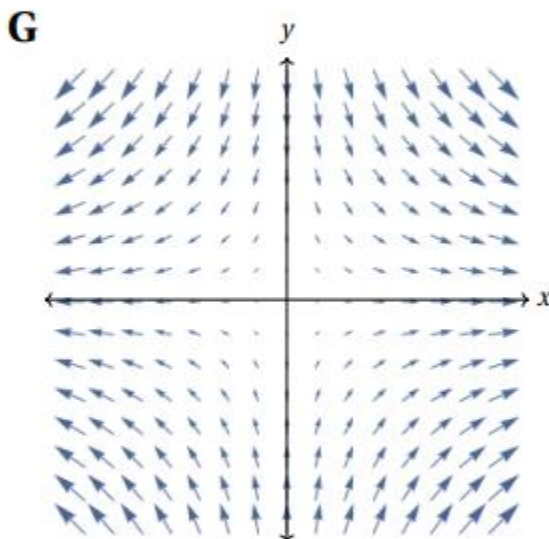
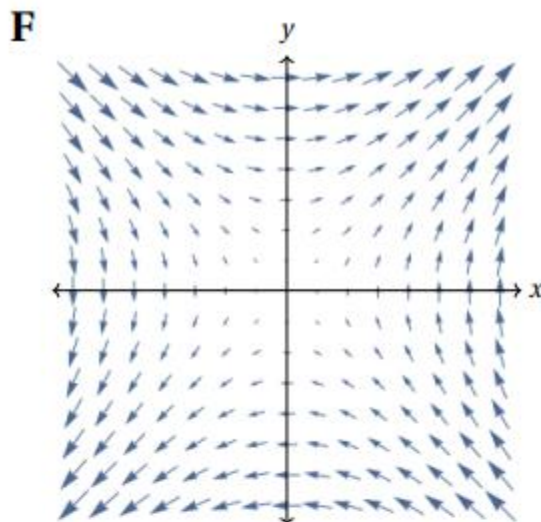
☐ positive ☐ negative ☒ zero

At every point on the line the direction of the vector field is orthogonal to the direction vector of the line and hence the work done is 0.

The work done by **H** on the particle is...

☒ positive ☐ negative ☐ zero

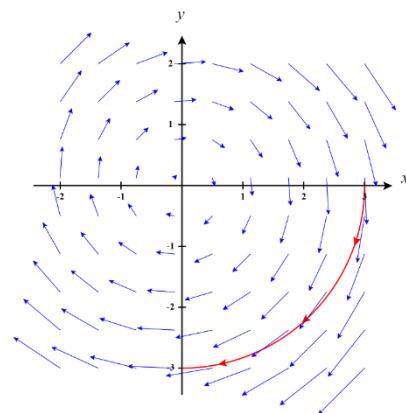
At every point on the line the direction of the vector field makes an angle of less than $\frac{\pi}{2}$ with the direction vector of the line. Therefore, the work done is positive.



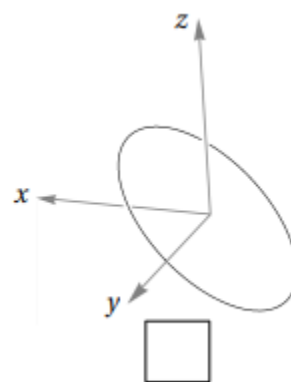
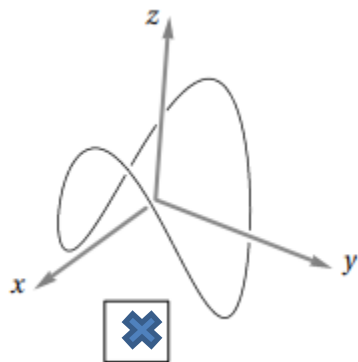
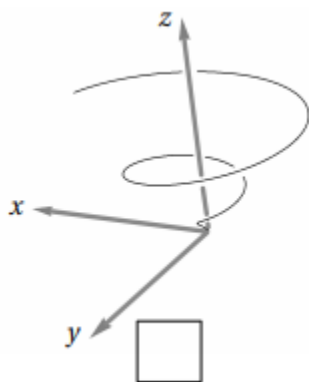
4. Find $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$ and C is the path along the circle of radius 3 starting at the point $(3,0)$ and ending at the point $(0,-3)$ while moving clockwise.

We begin by parameterizing the path C . One possible parameterization would be $\mathbf{r}(t) = \langle 3\cos(t), -3\sin(t) \rangle$ where $0 \leq t \leq \frac{\pi}{2}$.

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{\pi/2} \langle -3\sin(t), -3\cos(t) \rangle \cdot \langle -3\sin(t), -3\cos(t) \rangle dt \\ &= \int_0^{\pi/2} [9\sin^2(t) + 9\cos^2(t)] dt = \int_0^{\pi/2} 9 dt \\ &= \frac{9\pi}{2} \end{aligned}$$



5. Consider the curve C parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), \cos(2t) \rangle$ where $0 \leq t \leq 2\pi$.
- a) Check the box next to the correct sketch of C .



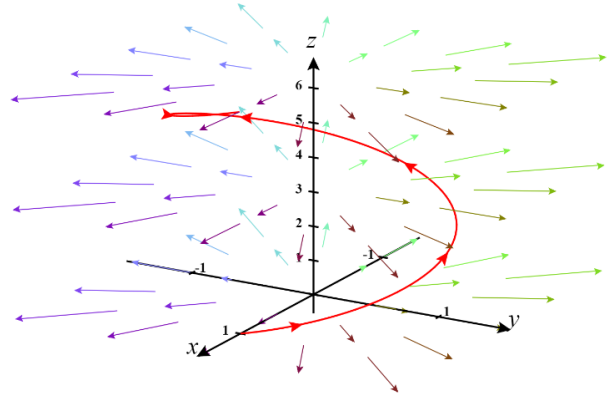
- b) Find the work done if a particle travels along path $\mathbf{r}(t)$ under the force given by the vector field $\mathbf{F}(x, y, z) = \langle -2y, 2x, 0 \rangle$.

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} \langle -2\sin(t), 2\cos(t), 0 \rangle \cdot \langle -\sin(t), \cos(t), -2\sin(2t) \rangle dt \\ &= \int_0^{2\pi} 2[\sin^2(t) + \cos^2(t)] dt \\ &= 4\pi \end{aligned}$$

6. Let C be the helix parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ where $0 \leq t \leq 2\pi$.

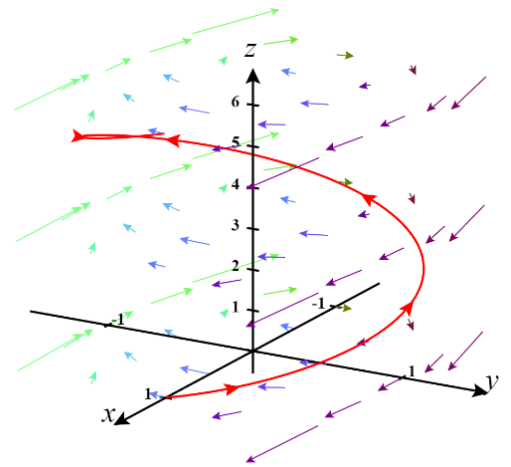
- a) Let $\mathbf{F}(x, y, z)$ be the vector field defined by $\mathbf{F}(x, y, z) = \langle x, y, 0 \rangle$. Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is oriented in the direction from $(1, 0, 0)$ to $(1, 0, 2\pi)$.

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} \langle \cos(t), \sin(t), 0 \rangle \cdot \langle -\sin(t), \cos(t), 1 \rangle dt \\ &= \int_0^{2\pi} [-\sin(t)\cos(t) + \sin(t)\cos(t) + 0] dt \\ &= \int_0^{2\pi} 0 dt \\ &= 0 \end{aligned}$$



- b) Now let $\mathbf{F}(x, y, z) = \langle ye^{x^2+y^2}, \ln(x^2+y^2) - x, 1 \rangle$. Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is given the opposite orientation.

$$\begin{aligned} \int_{-C} \mathbf{F} \cdot d\mathbf{r} &= - \int_C \mathbf{F} \cdot d\mathbf{r} \\ &= - \int_0^{2\pi} \langle \sin(t)e^1, \ln(1) - \cos(t), 1 \rangle \cdot \langle -\sin(t), \cos(t), 1 \rangle dt \\ &= - \int_0^{2\pi} [-\sin^2(t)e - \cos^2(t) + 1] dt \\ &= \int_0^{2\pi} [\sin^2(t)e + \cos^2(t) - 1] dt \\ &= \int_0^{2\pi} \left[\frac{e}{2} [1 - \cos(2t)] + \frac{1}{2} [1 + \cos(2t)] - 1 \right] dt \\ &= \left[\frac{e}{2} \left[t - \frac{1}{2} \sin(2t) \right] + \frac{1}{2} \left[t + \frac{1}{2} \sin(2t) \right] - t \right]_0^{2\pi} \\ &= \left[\frac{et}{2} - \frac{e}{4} \sin(2t) + \frac{t}{2} + \frac{1}{4} \sin(2t) - t \right]_0^{2\pi} \\ &= [\pi e - 0 + \pi + 0 - 2\pi] - [0 - 0 + 0 + 0 + 0] \\ &= \pi e - \pi \\ &= \pi(e - 1) \end{aligned}$$



7. Find the work done by the force field $\mathbf{F}(x, y) = \langle x^2, ye^x \rangle$ on a particle that moves along the parabola $x = y^2 + 1$ from $(1, 0)$ to $(2, 1)$ and then along the line segment from $(2, 1)$ to $(0, 0)$.

We note that we are moving along a curve C which is made up of two pieces.

Let C_1 be the portion of C that is determined by the parabola. This is parameterized by $\mathbf{r}_1(t) = \langle t^2 + 1, t \rangle$ where $0 \leq t \leq 1$.

Let C_2 be the portion of C that is determined by the line segment. This is parameterized by $\mathbf{r}_2(t) = \langle -2t + 2, -t + 1 \rangle$ where $0 \leq t \leq 1$.

We then calculate

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} \\ &= \int_0^1 \langle (t^2 + 1)^2, te^{t^2+1} \rangle \cdot \langle 2t, 1 \rangle dt + \int_0^1 \langle (-2t + 2)^2, (-t + 1)e^{-2t+2} \rangle \cdot \langle -2, -1 \rangle dt \\ &= \int_0^1 (2t^5 + 4t^3 + 2t + te^{t^2+1}) dt + \int_0^1 (-8t^2 + 16t - 8 - (-t + 1)e^{-2t+2}) dt \\ &= \int_0^1 (2t^5 + 4t^3 + 2t + te^{t^2+1}) dt + \int_0^1 (-8t^2 + 16t - 8 + te^{-2t+2} - e^{-2t+2}) dt \\ &= \left(\frac{1}{3}t^6 + t^4 + t^2 + \frac{1}{2}e^{t^2+1} \right) \Big|_0^1 + \left(\frac{8}{3}t^3 + 8t^2 - 8t + \frac{1}{2}e^{-2t+2} \right) \Big|_0^1 + \int_0^1 te^{-2t+2} dt \end{aligned}$$

We can use Integration by Parts with $u = t$ & $dv = e^{-2t+2}$ to evaluate the final integral.

Note that this implies that $du = dt$ & $v = -\frac{1}{2}e^{-2t+2}$.

$$\int_0^1 te^{-2t+2} dt = -\frac{1}{2}te^{-2t+2} \Big|_0^1 - \int_0^1 -\frac{1}{2}e^{-2t+2} dt$$

Therefore,

$$\begin{aligned} &\left(\frac{1}{3}t^6 + t^4 + t^2 + \frac{1}{2}e^{t^2+1} \right) \Big|_0^1 + \left(\frac{8}{3}t^3 + 8t^2 - 8t + \frac{1}{2}e^{-2t+2} \right) \Big|_0^1 + \int_0^1 te^{-2t+2} dt \\ &= \left(\frac{1}{3}t^6 + t^4 + t^2 + \frac{1}{2}e^{t^2+1} \right) \Big|_0^1 + \left(\frac{8}{3}t^3 + 8t^2 - 8t + \frac{1}{2}e^{-2t+2} \right) \Big|_0^1 + -\frac{1}{2}te^{-2t+2} \Big|_0^1 - \int_0^1 -\frac{1}{2}e^{-2t+2} dt \\ &= \left(\frac{1}{3} + 1 + 1 + \frac{1}{2}e^2 - \frac{1}{2}e \right) + \left(\frac{8}{3} + 8 - 8 + \frac{1}{2}e^0 - \frac{1}{2}e^2 \right) + \left(-\frac{1}{2}e^0 - 0 \right) - \left(\frac{1}{4}e^{-2t+2} \right) \Big|_0^1 \\ &= \left(\frac{7}{3} \right) + \left(\frac{8}{3} + \frac{1}{2} - \frac{1}{2}e^2 \right) - \frac{1}{2} - \left(\frac{1}{4}e^0 - \frac{1}{4}e^2 \right) \\ &= \frac{7}{3} + \frac{8}{3} - \frac{1}{2}e^2 - \frac{1}{4} + \frac{1}{4}e^2 \\ &= \frac{19}{4} - \frac{1}{4}e^2 \end{aligned}$$

8. Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y) = \langle x - y, xy \rangle$ and C is the arc of the circle $x^2 + y^2 = 4$ traversed counterclockwise from $(2, 0)$ to $(0, -2)$.

a) Evaluate this line integral in its current form using the formula $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \mathbf{r}'(t) dt$.

We first parameterize the C as $\mathbf{r}(t) = \langle 2 \cos(t), 2 \sin(t) \rangle$ where $0 \leq t \leq \frac{\pi}{2}$.

We then calculate

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{\pi/2} \langle 2 \cos(t) - 2 \sin(t), 4 \cos(t) \sin(t) \rangle \langle -2 \sin(t), 2 \cos(t) \rangle dt \\ &= \int_0^{\pi/2} (-4 \cos(t) \sin(t) + 4 \sin^2(t) + 8 \cos^2(t) \sin(t)) dt \\ &= \int_0^{\pi/2} -4 \cos(t) \sin(t) dt + \int_0^{\pi/2} 4 \sin^2(t) dt + \int_0^{\pi/2} 8 \cos^2(t) \sin(t) dt \\ &= -2 \sin^2(t) \Big|_0^{\pi/2} + \int_0^{\pi/2} 2[1 - \cos(2t)] dt - \frac{8}{3} \cos^2(t) \Big|_0^{\pi/2} \\ &= (-2 - 0) + 2 \left[t - \frac{1}{2} \sin(2t) \right] \Big|_0^{\pi/2} - \left[0 - \frac{8}{3} \right] = -2 + 2 \left[\frac{\pi}{2} - \frac{1}{2} \right] - \left[-\frac{8}{3} \right] = \pi - \frac{1}{3} \end{aligned}$$

b) Rewrite the integral in the form of $\int_C ? dx + ? dy$.

We note that $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \langle x - y, xy \rangle \cdot \langle dx, dy \rangle = \int_C (x - y) dx + xy dy$.

c) Evaluate the integral in the form of $\int_C ? dx + ? dy$.

We note that

$$\begin{aligned} \int_C (x - y) dx + xy dy &= \int_0^{\pi/2} (2 \cos(t) - 2 \sin(t))(-2 \sin(t)) dt + (4 \cos(t) \sin(t)) 2 \cos(t) dt \\ &= \int_0^{\pi/2} [(2 \cos(t) - 2 \sin(t))(-2 \sin(t)) + (4 \cos(t) \sin(t)) 2 \cos(t)] dt \\ &= \int_0^{\pi/2} (-4 \cos(t) \sin(t) + 4 \sin^2(t) + 8 \cos^2(t) \sin(t)) dt \\ &= \int_0^{\pi/2} -4 \cos(t) \sin(t) dt + \int_0^{\pi/2} 4 \sin^2(t) dt + \int_0^{\pi/2} 8 \cos^2(t) \sin(t) dt \\ &= -2 \sin^2(t) \Big|_0^{\pi/2} + \int_0^{\pi/2} 2[1 - \cos(2t)] dt - \frac{8}{3} \cos^2(t) \Big|_0^{\pi/2} \\ &= (-2 - 0) + 2 \left[t - \frac{1}{2} \sin(2t) \right] \Big|_0^{\pi/2} - \left[0 - \frac{8}{3} \right] = -2 + 2 \left[\frac{\pi}{2} - \frac{1}{2} \right] - \left[-\frac{8}{3} \right] = \pi - \frac{1}{3} \end{aligned}$$

9. Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y, z) = \langle \sin(x), \cos(y), xz \rangle$ and C is given by the vector function $\mathbf{r}(t) = t^3\mathbf{i} - t^2\mathbf{j} + t\mathbf{k}$ where $0 \leq t \leq 1$.

a) Evaluate this line integral in its current form using the formula $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \mathbf{r}'(t) dt$.

We calculate

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 \langle \sin(t^3), \cos(-t^2), (t^3)(t) \rangle \langle 3t^2, -2t, 1 \rangle dt = \int_0^1 (3t^2 \sin(t^3) - 2t \cos(-t^2) + t^4) dt \\ &= -\cos(t^3) + \sin(-t^2) + \frac{1}{5}t^5 \Big|_0^1 = \left(-\cos(1) + \sin(-1) + \frac{1}{5} \right) - (-1 + 0 + 0) \\ &= -\cos(1) + \sin(-1) + \frac{6}{5} \end{aligned}$$

b) Rewrite the integral in the form of $\int_C ? dx + ? dy + ? dz$.

$$\text{We note that } \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \langle \sin(x), \cos(y), xz \rangle \cdot \langle dx, dy, dz \rangle = \int_C \sin(x) dx + \cos(y) dy + xz dz.$$

c) Evaluate the integral in the form of $\int_C ? dx + ? dy + ? dz$.

We note that

$$\begin{aligned} \int_C \sin(x) dx + \cos(y) dy + xz dz &= \int_0^1 \sin(t^3) 3t^2 dt + \cos(-t^2) (-2t) dt + (t^3)(t) dt \\ &= \int_0^1 [3t^2 \sin(t^3) - 2t \cos(-t^2) + t^4] dt \\ &= -\cos(t^3) + \sin(-t^2) + \frac{1}{5}t^5 \Big|_0^1 \\ &= \left(-\cos(1) + \sin(-1) + \frac{1}{5} \right) - (-1 + 0 + 0) \\ &= -\cos(1) + \sin(-1) + \frac{6}{5} \end{aligned}$$

10. Consider the line integral $\int_C xy dx + 3y^2 dy$ where C is given by the vector function $\mathbf{r}(t) = \langle 11t^4, t^3 \rangle$ where $0 \leq t \leq 1$.

a) Evaluate this line integral in its current form.

$$\begin{aligned} \int_C xy dx + 3y^2 dy &= \int_0^1 (11t^4)(t^3)(44t^3) dt + 3(t^3)^2(3t^2) dt \\ &= \int_0^1 [11 \cdot 44t^{10} + 9t^8] dt = [44t^{11} + t^9]_0^1 \\ &= [44 + 1] - [0] \\ &= 45 \end{aligned}$$

b) Rewrite the integral in the form of $\int_C \mathbf{F} \cdot d\mathbf{r}$. What is $\mathbf{F}$? What is $d\mathbf{r}$?

$$\int_C xy dx + 3y^2 dy = \int_C \langle xy, 3y^2 \rangle \cdot \langle dx, dy \rangle = \int_C \mathbf{F} \cdot d\mathbf{r}$$

So we see that $\mathbf{F}(x, y) = \langle xy, 3y^2 \rangle$ and $d\mathbf{r} = \langle dx, dy \rangle$.

c) Evaluate the integral in the form of $\int_C \mathbf{F} \cdot d\mathbf{r}$ using the formula $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \mathbf{r}'(t) dt$.

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 \langle (11t^4)(t^3), 3(t^3)^2 \rangle \cdot \langle 44t^3, 3t^2 \rangle dt = \int_0^1 \langle 11t^7, 3t^6 \rangle \cdot \langle 44t^3, 3t^2 \rangle dt \\ &= \int_0^1 (44 \cdot 11t^{10} + 9t^8) dt = [44t^{11} + t^9]_0^1 \\ &= [44 + 1] - [0] \\ &= 45 \end{aligned}$$

11. Consider the integral $\int_C (x+y)dx + (y-z)dy + z^2dz$ where C is given by the vector function

$$\mathbf{r}(t) = \langle t^2, t^3, t^2 \rangle \text{ where } 0 \leq t \leq 1.$$

a) Evaluate this line integral in its current form.

$$\begin{aligned} \int_C (x+y)dx + (y-z)dy + z^2dz &= \int_0^1 (t^2 + t^3)(2t)dt + (t^3 - t^2)(3t^2)dt + (t^2)^2(2t)dt \\ &= \int_0^1 (2t^3 + 2t^4 + 3t^5 - 3t^4 + 2t^5)dt = \int_0^1 (2t^3 - t^4 + 5t^5)dt \\ &= \int_0^1 \left[\frac{1}{2}t^4 - \frac{1}{5}t^5 + \frac{5}{6}t^6 \right] dt \\ &= \left[\frac{1}{2} - \frac{1}{5} + \frac{5}{6} \right] - [0] = \left[\frac{15}{30} - \frac{6}{30} + \frac{25}{30} \right] \\ &= \frac{34}{30} = \frac{17}{15} \end{aligned}$$

b) Rewrite the integral in the form of $\int_C \mathbf{F} \cdot d\mathbf{r}$. What is $\mathbf{F}$? What is $d\mathbf{r}$?

$$\int_C (x+y)dx + (y-z)dy + z^2dz = \int_C \langle x+y, y-z, z^2 \rangle \cdot \langle dx, dy, dz \rangle = \int_C \mathbf{F} \cdot d\mathbf{r}$$

So we see that $\mathbf{F}(x, y, z) = \langle x+y, y-z, z^2 \rangle$ and $d\mathbf{r} = \langle dx, dy, dz \rangle$.

c) Evaluate the integral in the form of $\int_C \mathbf{F} \cdot d\mathbf{r}$ using the formula $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\mathbf{r}'(t)dt$.

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 \langle t^2 + t^3, t^3 - t^2, (t^2)^2 \rangle \cdot \langle 2t, 3t^2, 2t \rangle dt = \int_0^1 \langle t^2 + t^3, t^3 - t^2, t^4 \rangle \cdot \langle 2t, 3t^2, 2t \rangle dt \\ &= \int_0^1 [2t^3 + 2t^4 + 3t^5 - 3t^4 + 2t^5] dt = \int_0^1 (2t^3 - t^4 + 5t^5) dt \\ &= \int_0^1 \left[\frac{1}{2}t^4 - \frac{1}{5}t^5 + \frac{5}{6}t^6 \right] dt \\ &= \left[\frac{1}{2} - \frac{1}{5} + \frac{5}{6} \right] - [0] = \left[\frac{15}{30} - \frac{6}{30} + \frac{25}{30} \right] \\ &= \frac{34}{30} = \frac{17}{15} \end{aligned}$$